Chapter 13

How to work with forms

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Objectives

Applied

1. Use HTML to create a form that includes any of the form controls.
2. Use CSS to format a form and its controls.
3. Use the HTML attributes and CSS selectors for validating the data that gets entered into a form.
4. Use an HTML form to add a search function to a website.

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Objectives (continued)

Knowledge

1. Describe the use of an HTML form for initiating client-side processing or submitting data to a web server.
2. Describe the get and post methods that can be used to submit a form.
3. Describe the use of any of the form controls.
4. Describe the use of tab order and access keys.
5. Describe the use of these HTML attributes: autocomplete, required, and pattern.
6. Describe the use of regular expressions.
7. Describe the use of a form that provides a search function for a website.
8. Describe the use of a data list control.

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Attributes of the form element

```
id
action
method
```

Attributes common to most input elements

```
type
id
name
disabled
readonly
```


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The HTML for a form

```
<form id="email_form" action="subscribe.php"
      method="post">
    <p>Please enter your e-mail address to subscribe to
       our newsletter.</p>
    <p>E-Mail: <input type="text" id="email"></p>
    <p><input type="submit" id="submit"
              value="Subscribe"></p>
</form>
```


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The form in a web browser

Please enter your e-mail address to subscribe to our newsletter.

E-Mail:

Subscribe

The URL when the form is submitted with the get method

```
subscribe.php?email=zak%40modulemedia.com&submit=Subscribe
```


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Attributes of the input element for buttons and for the button element

```
type
value
src
alt
height
width
```


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Four buttons that are created by the input element

```
<input type="button" id="message" value="Alert Me">
<input type="submit" id="checkout" value="Checkout">
<input type="reset" id="resetform" value="Reset">
<input type="image" src="images/submit.jpg"
       alt="Submit button" width="114" height="42">
```

A button that is created by the button element

```
<button type="submit">
    <img src="images/addtocart.png" width="30"
         height="23" alt="Add to Cart">Add to Cart
</button>
```

The buttons in a web browser

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Attributes of the input element for text fields

```
type
value
maxlength
autofocus
placeholder
```


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The HTML for text fields

```
Username:<input type="text" id="username"
                autofocus><br><br>
Password:<input type="password"
                id="password"
                maxlength="6"
                placeholder="Enter your password">
<br><br>
Hidden:<input type="hidden" id="productid"
              value="widget">
```

The text fields in a web browser

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The HTML for a text area with default text

```
Comments:<br>
<textarea id="comments"
    placeholder="If you have any comments,
                 please enter them here.">
</textarea>
```

The CSS for the text area

```
textarea { height: 5em; width: 25em; }
```

The text area in a web browser

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The HTML for a form with label elements

```
Crust:<br>
<input type="radio" name="crust" id="crust1" value="thin"
       checked>
<label for="crust1">Thin Crust</label><br>
<input type="radio" name="crust" id="crust2"
       value="deep">
<label for="crust2">Deep Dish</label><br>
<input type="radio" name="crust" id="crust3"
       value="hand">
<label for="crust3">Hand Tossed</label><br><br>

Toppings:<br>
<input type="checkbox" name="topping1" id="topping1"
       value="pepperoni">
<label for="topping1">Pepperoni</label><br>
<input type="checkbox" name="topping2" id="topping2"
       value="mushrooms">
<label for="topping2">Mushrooms</label><br>
<input type="checkbox" name="topping3" id="topping3"
       value="Black Olives">
<label for="topping3">Black Olives</label><br><br>
```


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The radio buttons and check boxes in a web browser

Crust:

- (●) Thin Crust
- ( ) Deep Dish
- ( ) Hand Tossed

Toppings:

- [ ] Pepperoni
- [ ] Mushrooms
- [ ] Black Olives

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Accessibility guideline for labels with radio buttons and check boxes

- Use labels with radio buttons and check boxes so the user can click on the label to select the control that the label is associated with.
- This also helps assistive devices.

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The HTML for a drop-down list

```
Style:<br>
<select id="style_and_size">
    <optgroup label="The New Yorker">
        <option value="ny10">10"</option>
        <option value="ny12">12"</option>
        <option value="ny16">16"</option>
    </optgroup>
    <optgroup label="The Chicago">
        <option value="chi10">10"</option>
        <option value="chi12">12"</option>
        <option value="chi16">16"</option>
    </optgroup>
</select>
```


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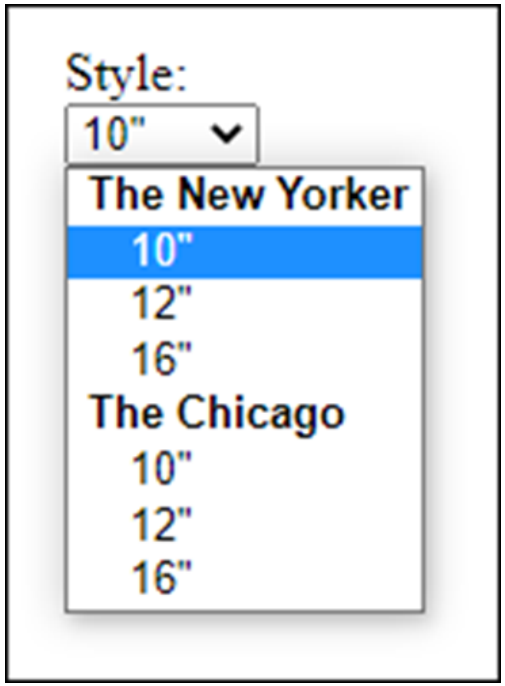
The drop-down list in a web browser when the user clicks on the arrow



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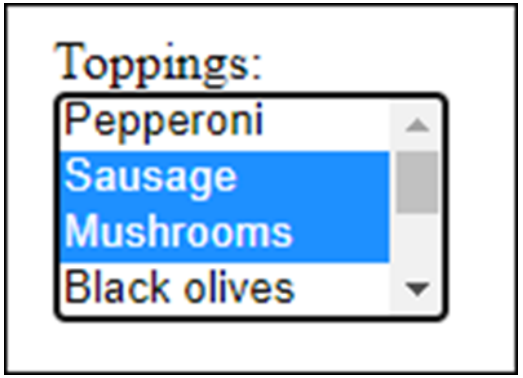
The HTML for a list box

```
<select id="toppings" size="4" multiple>
  <option value="pepperoni">Pepperoni</option>
  <option value="sausage" selected>Sausage</option>
  <option value="mushrooms">Mushrooms</option>
  <option value="olives">Black olives</option>
  ...
</select>
```

The list box in a web browser with a scroll bar



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HTML code that uses the number, email, url, and tel controls

```
<form id="email_form" action="survey.php" method="post">
    <label for="investment">Monthly investment: </label>
    <input type="number" id="investment"
           min="100" max="1000" step="100"
           value="300"><br><br>
    <label for="email">Your email address:</label>
    <input type="email" id="email" required><br>
    <label for="link">Your website:</label>
    <input type="url" id="link" list="links"><br>
    <label for="phone">Your phone number:</label>
    <input type="tel" id="phone" required><br><br>
    <input type="submit" id="submit"
           value="Submit Survey">
</form>
```


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The form in Chrome

Monthly investment: 300

Your email address: zak@modulemedia.com

Your website:

Your phone number:

Submit Survey

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HTML that uses the date and time controls

```
Date and time:  
    <input type="datetime" name="datetime"><br><br>
Local date and time:  
    <input type="datetime-local"
           id="datetimelocal"><br><br>
Month:  
    <input type="month" id="month"><br><br>
Week:  
    <input type="week" id="week"><br><br>
Time:  
    <input type="time" id="time"><br><br>
Date:  
    <input type="date" id="date">
```


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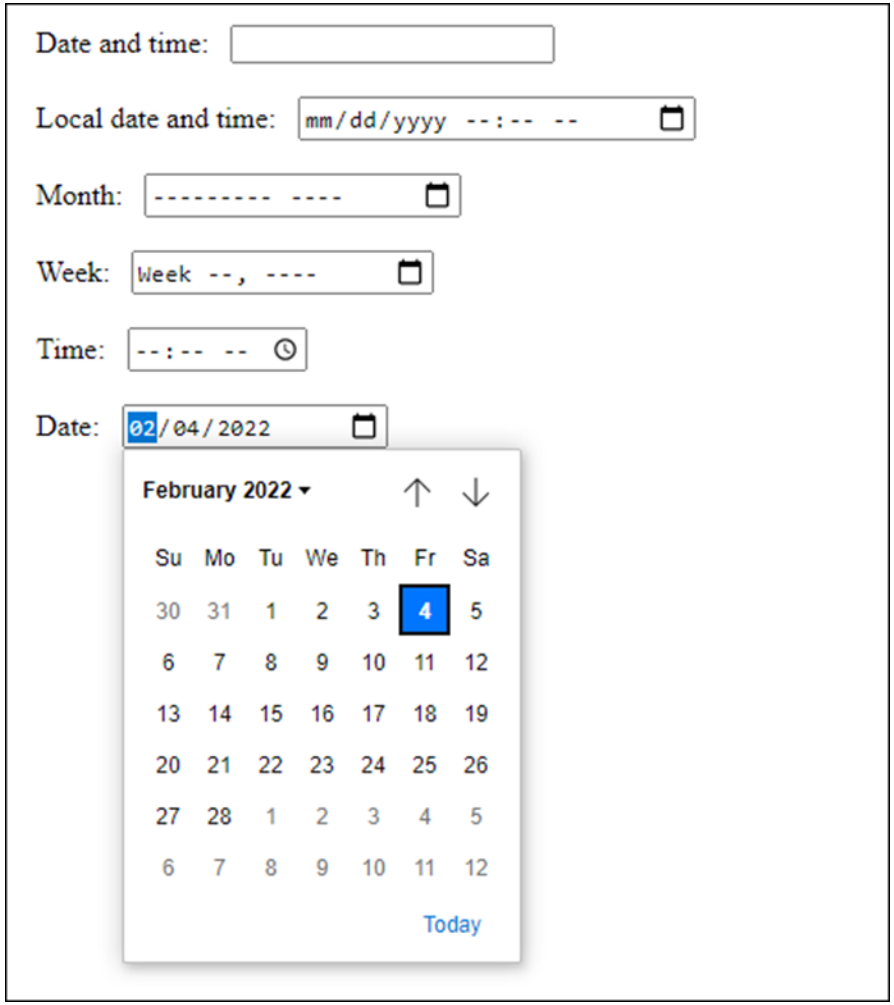
The controls in Chrome



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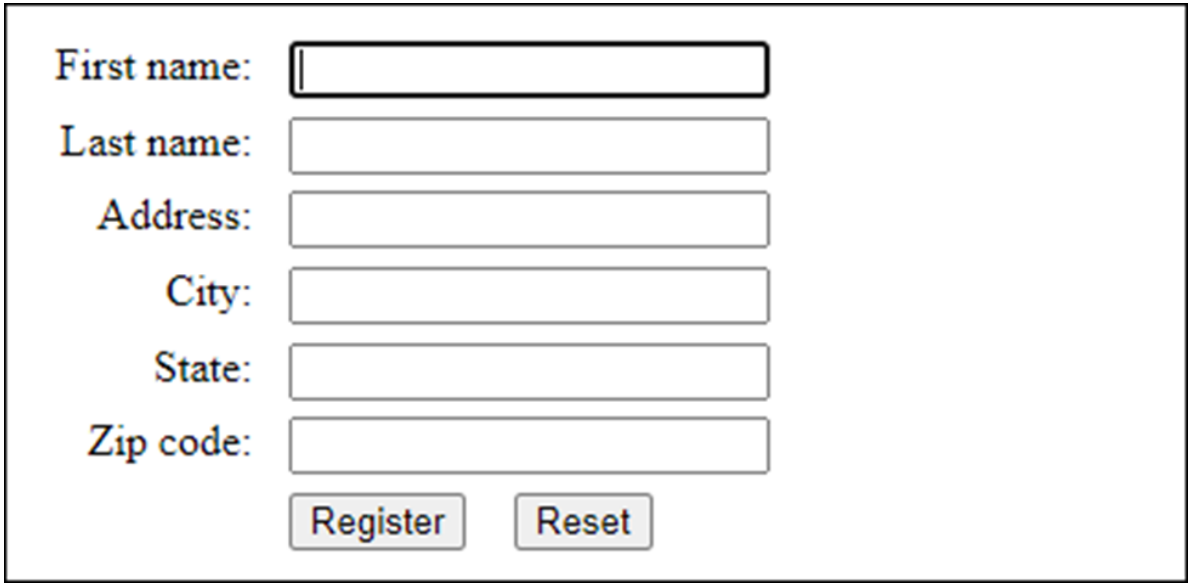
Label, text box, and button controls aligned on a form



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The HTML for the form

```
<label for="firstname">First name:</label>
<input type="text" id="firstname" autofocus><br>
<label for="lastname">Last name:</label>
<input type="text" id="lastname"><br>
<label for="address">Address:</label>
<input type="text" id="address"><br>
<label for="city">City:</label>
<input type="text" id="city"><br>
<label for="state">State:</label>
<input type="text" id="state"><br>
<label for="zip">Zip code:</label>
<input type="text" id="zip"><br>
<input type="submit" id="button" value="Register">
<input type="reset" name="reset" id="reset">
```


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The CSS for the controls

```
label {
    float: left;
    width: 5em;
    text-align: right;}
input {
    margin-left: 1em;
    margin-bottom: .5em;}
#button {
    margin-left: 7em;}
```


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HTML that uses fieldset and legend elements

```
<form id="order" action="order.php" method="post">
<fieldset>
    <legend>Crust</legend>
    <input type="radio" name="crust" id="crust1"
           value="thin">
    <label for="crust1">Thin Crust</label><br>
    <input type="radio" name="crust" id="crust2"
           value="deep">
    <label for="crust2">Deep Dish</label><br>
    ...
</fieldset>
<br>
<fieldset>
    <legend>Toppings</legend>
    <input type="checkbox" name="topping1" id="topping1"
           value="pepperoni">
    <label for="topping1">Pepperoni</label><br>
    ...
</fieldset>
</form>
```


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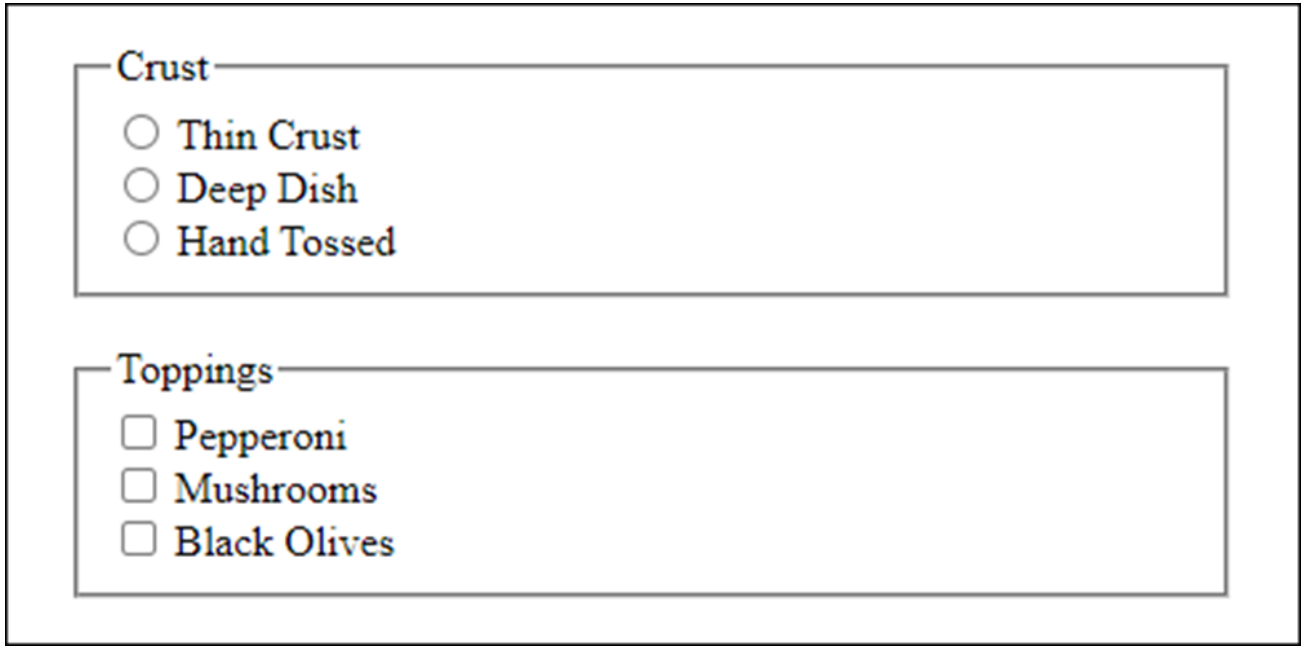
The elements in a web browser



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Three labels with access keys

First name:

Last name:

Email:

The HTML for the controls

```
<label for="firstname"><u>F</u>irst name:</label>
<input type="text" id="firstname" accesskey="F"><br>
<label for="lastname"><u>L</u>ast name:</label>
<input type="text" id="lastname" accesskey="L"><br>
<label for="email"><u>E</u>mail:</label>
<input type="text" id="email" accesskey="E">
```


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Another way to define the access keys

```
<label for="firstname" accesskey="F"><u>F</u>irst name:
</label>
<input type="text" id="firstname"><br>
<label for="lastname" accesskey="L"><u>L</u>ast name:
</label>
<input type="text" id="lastname"><br>
<label for="email" accesskey="E"><u>E</u>mail:</label>
<input type="text" id="email">
```


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Accessibility guideline for tab order and access keys

- Setting a proper tab order and providing access keys improves the accessibility for users who can't use a mouse.

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The HTML attributes for data validation

```
autocomplete
required
novalidate
formnovalidate
```


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HTML that uses the validation attributes

```
Name:    <input type="text" name="name" required><br>
Address: <input type="text" name="address"><br>
Zip:     <input type="text" name="zip" required><br>
Phone:   <input type="text" name="phone" required
               autocomplete="off"><br>
<input type="submit" name="submit" value="Submit Survey">
```


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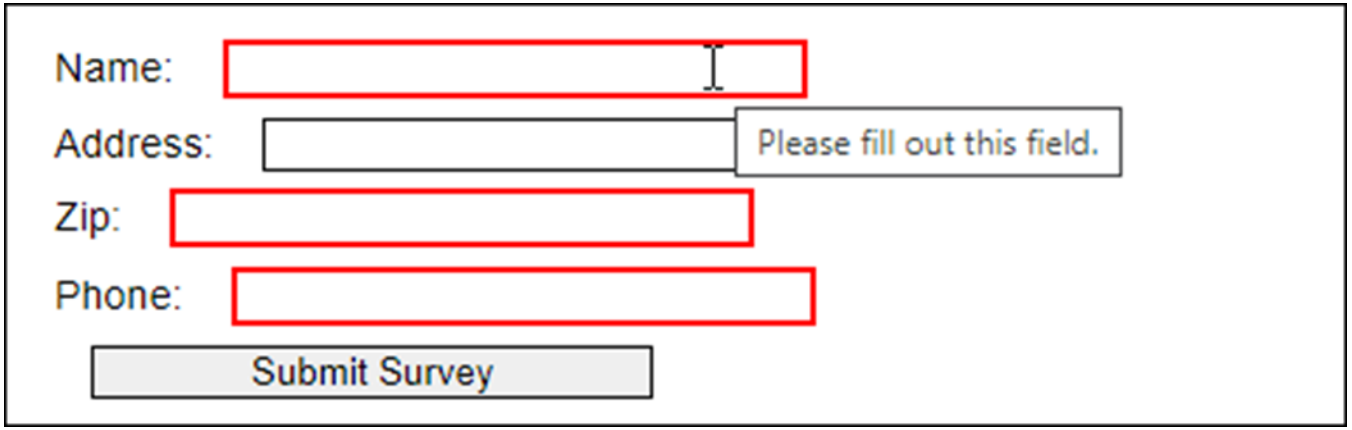
The error message and highlighting in Chrome



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The CSS pseudo-classes for required, valid, and invalid fields

```
:required
:valid
:invalid
```


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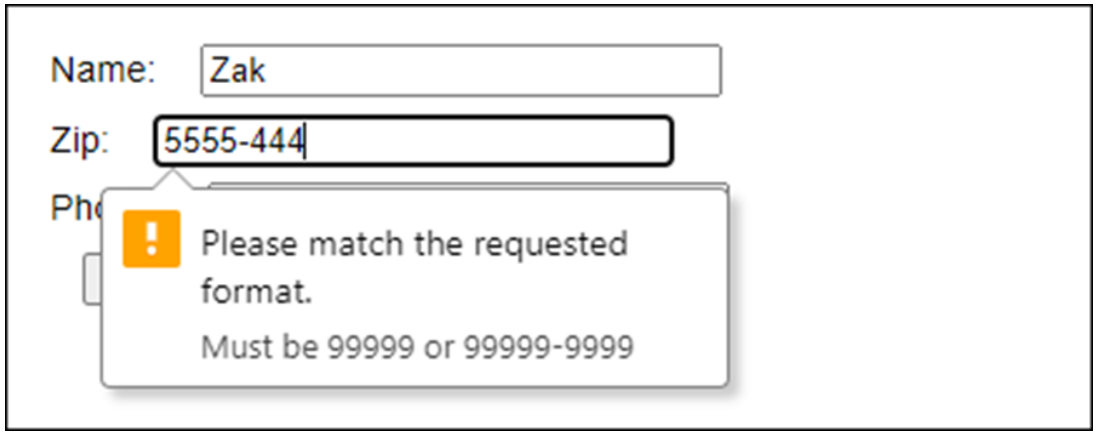
HTML that uses regular expressions

```
Name: <input type="text" id="name"
             required autofocus><br>
Zip: <input type="text" id="zip" required
            pattern="\d{5}([\-]\d{4})?"
            title="Must be 99999 or 99999-9999"><br>
Phone: <input type="text" id="phone" required
              pattern="\d{3}[\-]\d{3}[\-]\d{4}"
              title="Must be 999-999-9999"><br>
<input type="submit" id="submit" value="Submit Survey">
```

The form in Chrome



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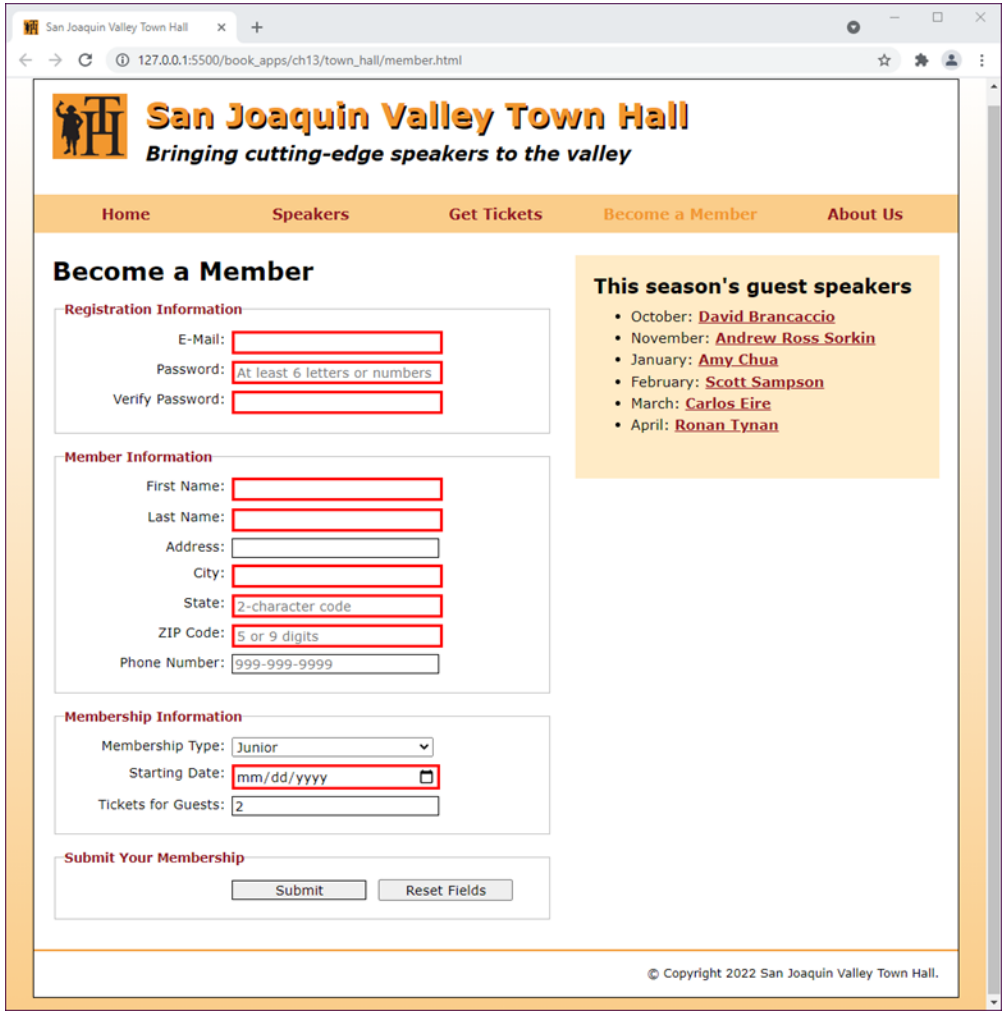
A form in Chrome that uses HTML validation



The HTML for the form (part 1)

```
<form action="register_account.html" method="get"
      id="registration_form">
<fieldset>
    <legend>Registration Information</legend>
    <label for="email">E-Mail:</label>
    <input type="email" name="email" id="email"
           autofocus required><br>
    <label for="password">Password:</label>
    <input type="password" name="password" id="password" required
           placeholder="At least 6 letters or numbers"
           pattern="[a-zA-Z0-9]{6,}"
           title="Must be at least 6 alphanumeric characters">
    <br>
    <label for="verify">Verify Password:</label>
    <input type="password" name="verify"
           id="verify" required><br>
</fieldset>
```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The HTML for the form (part 2)

```
<fieldset>
    <legend>Member Information</legend>
    <label for="first_name">First Name:</label>
    <input type="text" name="first_name" id="first_name"
           required><br>
    ...
    ...
    ...
    <label for="state">State:</label>
    <input type="text" name="state" id="state" required
           maxlength="2"
           placeholder="2-character code"><br>
    <label for="zip">ZIP Code:</label>
    <input type="text" name="zip" id="zip" required
           placeholder="5 or 9 digits"
           pattern="^\d{5}(-\d{4})?$"
           title="Either 5 or 9 digits"><br>
    <label for="phone">Phone Number:</label>
    <input type="tel" name="phone" id="phone"
           placeholder="999-999-9999"
           pattern="\d{3}[\-]\d{3}[\-]\d{4}"
           title="Must be 999-999-999 format"><br>
</fieldset>
```


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The HTML for the form (part 3)

```
<fieldset>
    <legend>Membership Information</legend>
    <label for="membership_type">Membership Type:</label>
    <select name="membership_type" id="membership_type">
        <option value="j">Junior</option>
        <option value="r">Regular</option>
        <option value="c">Charter</option>
    </select><br>
    <label for="starting_date">Starting Date:</label>
    <input type="date" name="starting_date" id="starting_date"
           required><br>
    <label for="tickets">Tickets for Guests:</label>
    <input type="number" name="tickets" id="tickets"
           value="2" min="1" max="4"
           placeholder="from 1 to 4"><br>
</fieldset>
<fieldset id="buttons">
    <legend>Submit Your Membership</legend>
    <label> </label>
    <input type="submit" id="submit" value="Submit">
    <input type="reset" id="reset" value="Reset Fields"><br>
</fieldset>
</form>
```


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The CSS for the form

```
fieldset {
    margin-top: 1em;
    margin-bottom: 1em;
    padding-top: .5em;
}
legend {
    color: #931420;
    font-weight: bold;
    font-size: 85%;
    margin-bottom: .5em;
}
label, input, select {
    font-size: 90%;
}
label {
    float: left;
    width: 12em;
    text-align: right;
}
```


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The CSS for the form (continued)

```
input, select {
    width: 15em;
    margin-left: .5em;
    margin-bottom: .5em;
}
input:required {
    border: 3px solid red;
}
input:valid {
    border: 1px solid black;
}
input:invalid {
    box-shadow: none;
}
br {
    clear: both;
}
#buttons input {
    width: 10em;
}
```


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A search control in the Chrome browser

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The results of the search when Google is used

Google

sql server site:http://www.murach.com

All | Videos | News | Images | Books | More | Tools

About 74 results (0.39 seconds)

https://www.murach.com › shop › murach-s-sql-server-...

Murach's SQL Server 2019 for Developers

We've heard from a lot of experienced developers who read our programming book, **SQL Server** for Developers, and were amazed to discover how much they didn't ...

http://www.murach.com › instructors › shop › murach-s-s...

Murach's SQL Server 2019 for Developers

This book presents all of the **SQL Server** 2019 skills that application developers need on the job. As such, it's a great book for a SQL course.

http://www.murach.com › instructors › shop › murach-...

Murach's SQL Server 2016 for Developers

This book presents all of the **SQL Server** skills that application developers need on ... **SQL Server** Programming Book for College Instructors | Murach Books ...

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The HTML for using the Google search engine

```
<form method="get" action="http://www.google.com/search">
     <input type="search" name="q" size="30"
            maxlength="255">
     <input type="hidden" name="domains"
            value="http://www.murach.com">
     <input type="hidden" name="sitesearch"
            value="http://www.murach.com">
     <input type="submit" id="search" value="Search">
</form>
```


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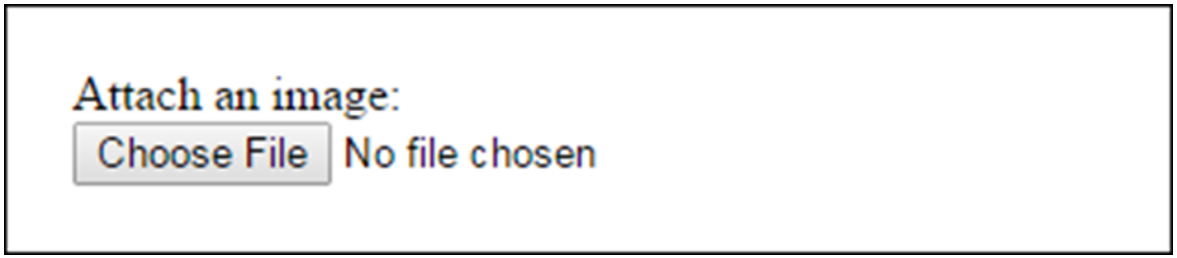
The HTML for a file upload control that accepts JPEG and GIF images

```
<form id="upload_form" action="sendemail.php"
      method="post"
      enctype="multipart/form-data">
    Attach an image:<br>
    <input type="file" id="fileupload"
           accept="image/jpeg, image/gif">
</form>
```

The file upload control in the Chrome browser



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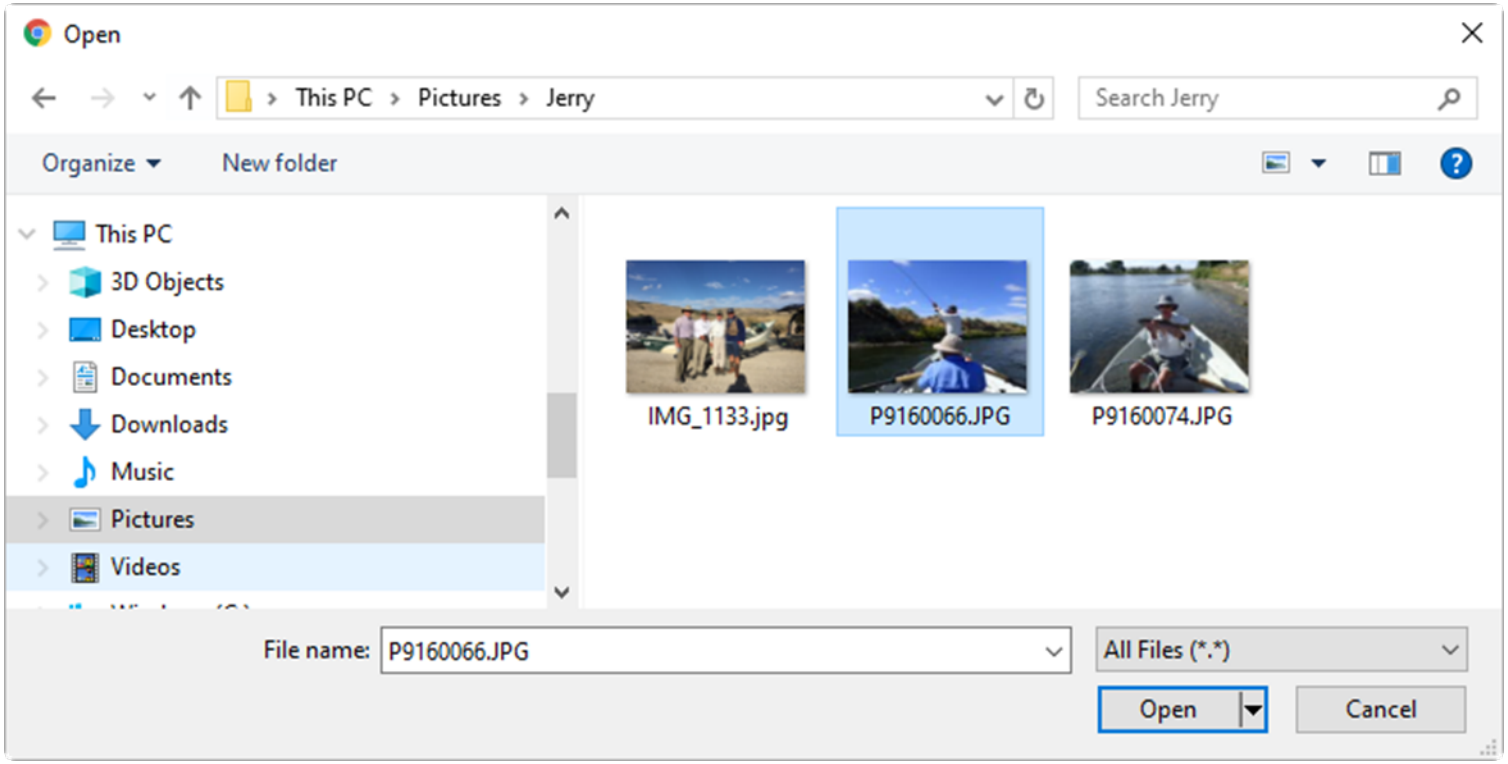
The Windows dialog box when Choose File is clicked



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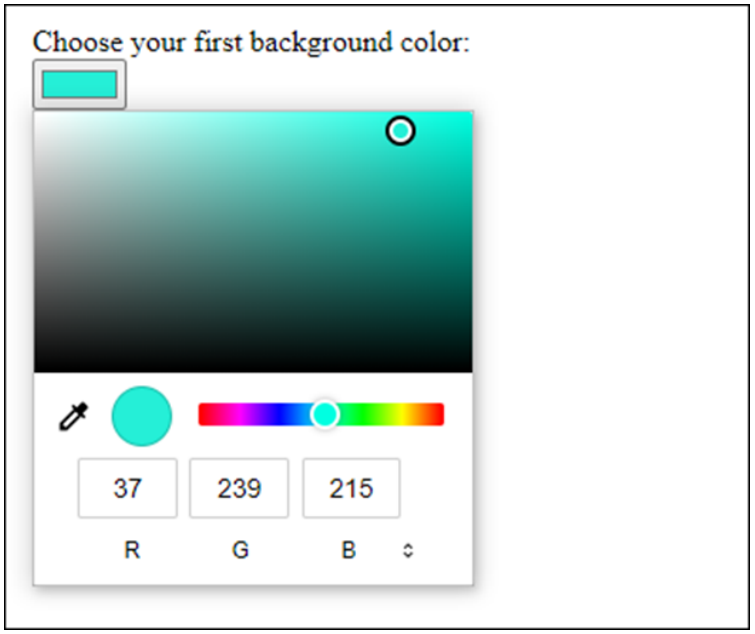
The HTML for a color control

```
<label for="firstcolor">
    Choose your first background color:
</label>
<input type="color" id="firstcolor" value="#facd8a">
```

The color control in Chrome



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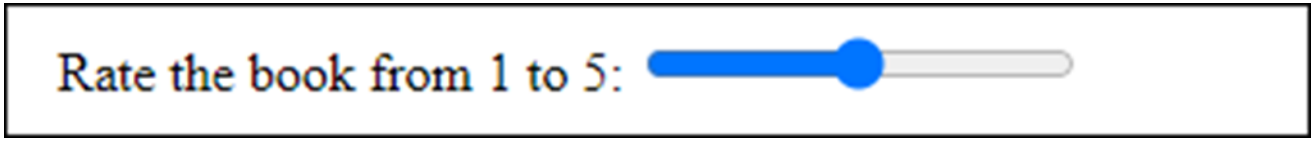
The HTML for a range control

```
<label for="book">Rate the book from 1 to 5: </label>
<input type="range" id="book"
       min="1" max="5" step="1"><br><br>
```

The range control in Chrome



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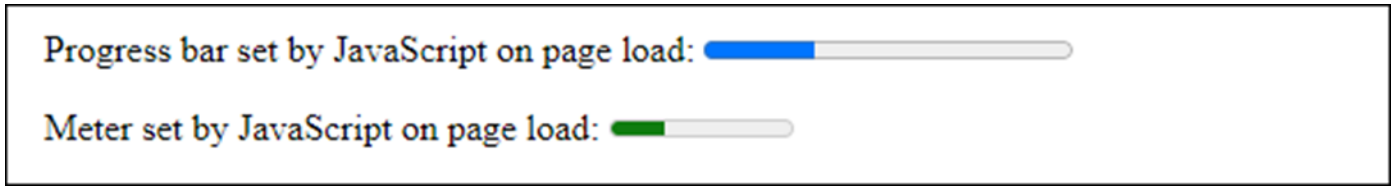
The HTML for progress and meter controls

```
<body onload="setProgressAndMeter()">
    <p>Progress bar set by JavaScript on page load:
        <progress id="progressBar" max="100"
                  value="0"></progress></p>
    <p>Meter set by JavaScript on page load:
        <meter id="meterBar" max="100"
               value="0"></meter></p>
</body>
```

The progress and meter controls in Chrome



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The HTML for a datalist control

```
<label for="link">What is your preferred search engine:
</label>
<input type="url" name="link" id="link" list="links">
<br><br>
<datalist id="links">
    <option value="http://www.google.com/"
            label="Google"></option>
    <option value="http://www.yahoo.com/"
            label="Yahoo"></option>
</datalist>
```


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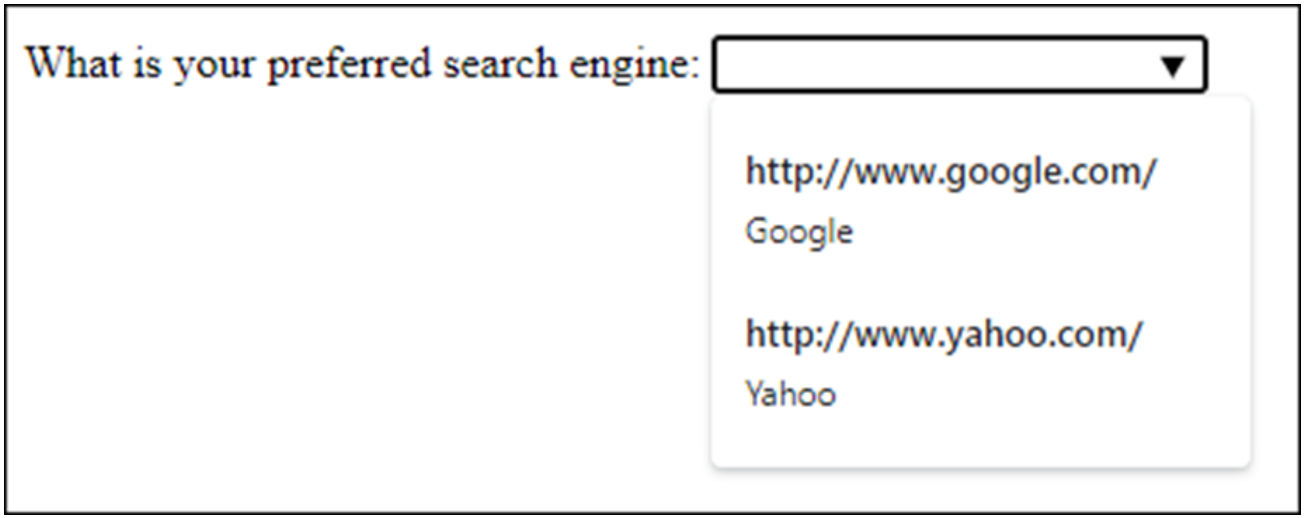
The datalist in Chrome



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The HTML for an output control

```
<p>Enter numbers in both fields and click the Calculate
   button.</p>
<form onsubmit="return false">
    <input name="x" type="number" min="100" step="5"
           value="100"> +
    <input name="y" type="number" min="100" step="5"
           value="100"><br><br>
    <input type="button" value="Calculate"
        onclick="result.value = parseInt(x.value) +
                 parseInt(y.value)">
    <br><br>
    Total: <output name="result" for="x y"></output>
</form>
```


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The form in Chrome with a border around the output element

Enter numbers in both fields and click the Calculate button.

100 + 100

Calculate

Total: 200

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Short 13-1 Create a form

Please complete our survey

Contact information

Email address:

First name:

Last name:

Geographic information

State: Two characters

Zip code: Five digits

How did you hear about us?

Web Search:

Facebook:

Twitter:

Email message:

Thank you for taking our survey!

Submit Reset

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